Here, $P(A) = \frac{18}{36} = \frac{1}{2}$

$P(B) = \frac{10}{36} = \frac{5}{9}$

$P(C) = \frac{30}{36} = \frac{5}{6}$.

Since, $A \cap C = A$. Hence,

$P(A \cap C) = P(A) = \frac{1}{2}$

$P(B \cap C) = P(B) = \frac{5}{9}$.

Hence, the probability for the sum of the faces is an odd number given that the faces are different is

$$\mathrm{P(A/C)} = \frac{P(A \cap C)}{P(C)} = \frac{\frac{1}{2}}{\frac{5}{6}} = \frac{3}{5}$$

Similarly, the probability for the sum of the faces exceeds 8 given that the faces are different is

$$\mathrm{P(B/C)} = \frac{P(B \cap C)}{P(C)} = \frac{\frac{2}{9}}{\frac{5}{6}} = \frac{4}{15}$$

8.5.2 Axioms

The conditional probabilities also satisfy the same axioms introduced in Section 8.3.

If S is the sample space of a random experiment and B is an event in the experiment, then

(i) $\mathrm{P(A/B)} \geq 0$ for any event A of S.

(ii) $\mathrm{P(S/B)} = 1$

(iii) If $A_1, A_2, \ldots$ is a sequence of mutually exclusive events, then

$$P\left(\bigcup_{i=1}^{\infty} A_i \mid B\right) = \sum_{i=1}^{\infty} P(A_i \mid B)$$

In continuation of conditional probability, another property of events, viz., independence can be studied. It is discussed in the next section. Also, multiplication theorem, a consequence of conditional probability, will be studied later.

8.6 Independent Events

For any two events A and B of a random experiment, if $P(A/B) = P(A)$, then knowledge of the event B does not change the probability for the occurrence of the event A. Such events are called independent events.